

DS 301 FINAL PROJECT REPORT

Credit Card Default Prediction

Reproduction of Three Supervised Models and a Logistic Regression Contribution

Team Members

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Scientific basis

Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms

UCI Default of Credit Card Clients dataset

1. Summary

This project investigates whether a credit card client will default on the next monthly payment. It is based only on the selected paper Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms, which experimentally compares Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) using the UCI Default of Credit Card Clients dataset.

The group reproduced the paper's main workflow by removing undocumented categories, creating a stratified 80/20 train-test split, standardizing the features, applying random oversampling only to the training data, and training the three reported models with the paper's best parameters. The reproduced results were close to the published results and confirmed SVM as the strongest balanced model.

The main group contribution was adding Logistic Regression as a fourth classifier. The model was tuned with three-fold stratified cross-validation over five C values and selected using default-class F1-score. Logistic Regression achieved the highest recall in the group experiment, 0.645, but its F1-score of 0.486 remained below the reproduced SVM result of 0.510. Therefore, the contribution improved the detection of actual default cases but did not improve the best overall F1-score.

2. Motivation

Credit default prediction is an important classification problem because banks need to identify clients who may miss future payments. False negatives can expose the institution to financial risk, while false positives can lead to unnecessary interventions or unfair treatment of safe clients. For this reason, model evaluation should consider precision, recall, and F1-score rather than accuracy alone.

The paper was selected because it addresses a practical financial problem, uses a public and reproducible dataset, compares three models covered in the course, and provides clear opportunities for an additional classification experiment. The project focused on two questions:

- Can the main workflow and results of the selected paper be reproduced?
- What additional insight does Logistic Regression provide when evaluated on the same cleaned data and test set?

3. Research Paper Details

The selected paper was written by Thomson Ly, Rene Schuller, Katarina Simanic, and Sukhpreet Singh at the University of Windsor. It studies credit card default as a supervised binary classification problem and evaluates Decision Tree, KNN, and SVM with accuracy, precision, recall, F1-score, and confusion matrices.

Model	Best parameters reported	Accuracy	Default F1
Decision Tree	entropy; max depth=None; max features=9; splitter=best	0.740	0.410
KNN	neighbors=1; metric=euclidean; weights=uniform	0.730	0.390
SVM	kernel=RBF; C=10	0.760	0.520

The article concluded that SVM was the best of the three tested models. However, its default-class F1-score was still moderate, so the authors did not claim that the models provided high confidence for real credit decisions.

4. Dataset Details

The project uses the Default of Credit Card Clients dataset from the UCI Machine Learning Repository, dataset ID 350. The records describe credit card clients from Taiwan and contain demographic, credit, payment-status, bill, and previous-payment information. The target indicates whether the client defaults in the following month.

Item	Value
Original observations	30,000
Removed observations	399
Final observations	29,601
Input features	23
No-default clients	22,996 (77.7%)
Default clients	6,605 (22.3%)
Target	Default next month: 0 or 1

4.1 Feature Groups and Data Types

Group	Columns	Analytical type
Credit limit	LIMIT_BAL	Numerical
Demographics	SEX, EDUCATION, MARRIAGE	Integer-coded categorical
Age	AGE	Numerical
Payment status	PAY_0, PAY_2-PAY_6	Ordered integer status
Bill amounts	BILL_AMT1-BILL_AMT6	Numerical
Payment amounts	PAY_AMT1-PAY_AMT6	Numerical
Target	DEFAULT	Binary class

The source data has no missing values. However, EDUCATION and MARRIAGE contain category values not documented by the dataset description. Following the paper, the project removed those rows. The target is imbalanced because only 22.3% of the remaining clients belong to the default class.

5. Data Preprocessing

1. Load the local Excel dataset; use UCI dataset ID 350 as an online fallback.
2. Remove the ID column because it is only an identifier.
3. Keep documented EDUCATION values 1-4 and MARRIAGE values 1-3, removing 399 rows.
4. Create a stratified 80% training and 20% testing split using random state 42.
5. Fit StandardScaler on the training data and transform both training and test features.
6. Apply random oversampling only to the scaled training set, balancing it at 18,396 records per class.
7. Keep the test set unchanged for final evaluation.

5.1 Feature Engineering and Critical Observations

No synthetic features were required. The main preparation decisions were removing undocumented categories, standardizing the 23 inputs, and balancing the training classes. Oversampling was applied only after the split and only to the training data, which kept the test distribution realistic and reduced data leakage risk.

6. Steps Reproduced from the Paper

The project reproduces the paper's cleaning, split, scaling, balancing, model training, and evaluation stages. For a short and explainable classroom notebook, the reproduced Decision Tree, KNN, and SVM use the best parameters reported by the paper instead of repeating the entire article grid search. A fixed random state of 42 was added because the paper does not provide its original split seed.

Step	Reproduced implementation
Cleaning	Remove ID and undocumented EDUCATION or MARRIAGE rows.
Split	Stratified 80/20 train-test split.
Scaling	StandardScaler fitted only on training data.
Balancing	Random oversampling applied only to training data.
Models	Decision Tree, KNN, and SVM with reported best parameters.
Evaluation	Accuracy, precision, recall, F1-score, and confusion matrix.

6.1 Reproduction Results

Model	Accuracy	Precision	Recall	F1	Article F1	Difference
Decision Tree	0.735	0.404	0.399	0.401	0.410	-0.009
KNN	0.723	0.379	0.377	0.378	0.390	-0.012
SVM	0.757	0.464	0.566	0.510	0.520	-0.010

All reproduced F1-scores are within 0.012 of the published values. This supports the conclusion that the main paper workflow was reproduced successfully. SVM remained the strongest of the three models.

7. Contributions

Our principal contribution was implementing, tuning, and evaluating Logistic Regression. This model was not experimentally compared among the paper's three main approaches.

Contribution element	Implementation and purpose
Additional model	Logistic Regression added as a fourth binary classifier.
Parameter tuning	Five C values tested: 0.01, 0.1, 1, 10, and 100.
Validation	Three-fold stratified cross-validation.
Selection metric	Default-class F1-score used to select the best C.
Fair comparison	Same cleaned data, balanced training set, and untouched test set.
Interpretation	Performance compared directly with the reproduced SVM.

The best configuration used $C=0.01$ with the liblinear solver. Logistic Regression is appropriate because the target is binary. Linear Regression was not used because it predicts continuous numerical outcomes rather than binary classes.

8. Significant Improvements

The contribution did not improve the best overall F1-score. Instead, it produced a meaningful improvement in recall. This distinction is important because the assignment requires an evidence-based argument about whether the contribution improved performance.

Metric	Reproduced SVM	Logistic Regression	Change with Logistic Regression
Accuracy	0.757	0.696	-0.062
Precision	0.464	0.390	-0.074
Recall	0.566	0.645	+0.079
F1-score	0.510	0.486	-0.024

Main interpretation: Logistic Regression detected 64.5% of the clients who actually defaulted, compared with 56.6% for the reproduced SVM. However, it also classified more safe clients as risky, reducing precision, accuracy, and F1-score. SVM remains the better balanced model, while Logistic Regression may be useful when the business priority is to miss fewer high-risk clients.

8.1 Logistic Regression Confusion Matrix

Actual class	Predicted: No default	Predicted: Default
Actual: No default	3,267	1,333
Actual: Default	469	852

The model correctly detected 852 default clients and missed 469. The 1,333 false positives explain why recall increased while precision remained lower.

9. Challenges

- Class imbalance: only 22.3% of the cleaned records belong to the default class.
- Exact reproduction: the paper does not provide its original random split seed or every implementation detail.
- Computation time: reproduced SVM required substantially more training time than Decision Tree, KNN, or Logistic Regression.
- Metric trade-offs: increasing recall can reduce precision and create more false-positive decisions.
- Business interpretation: the model with the highest recall is not automatically the strongest overall model.

9.1 Main Learnings

- Accuracy alone is incomplete for an imbalanced classification problem.
- Recall measures the proportion of actual default clients detected by the model.
- F1-score is useful when both precision and recall matter.
- Preprocessing and oversampling must be learned from training data, not the test set.
- A contribution can add business value even if it does not produce the highest F1-score.
- Clear documentation and fixed random states make scientific reproduction easier to explain.

10. Conclusion and Future Scope

10.1 Conclusion

The project successfully reproduced the main workflow and conclusion of the selected paper. The reproduced F1-scores were close to the published values, and SVM remained the strongest balanced model with a reproduced F1-score of 0.510.

Logistic Regression was implemented as the group's additional classification model. It achieved accuracy 0.696, precision 0.390, recall 0.645, and F1-score 0.486. It did not outperform SVM on F1-score, but it increased recall by 0.079. The result demonstrates that model selection depends on the business objective: SVM is preferable for balance, while Logistic Regression can be useful when missing fewer actual defaulters is the priority.

10.2 Limitations

- The dataset represents clients from one Taiwanese bank and an older historical period.
- Random oversampling repeats minority-class observations and may increase overfitting.
- The article does not provide every detail required for an identical reproduction.
- The project does not include probability calibration or a financial cost matrix.
- The models should not be used alone to make automatic credit decisions.

10.3 Future Scope

- Test additional models such as Random Forest, XGBoost, and neural networks.
- Perform a broader hyperparameter search using cross-validation.
- Compare imbalance techniques such as SMOTE, class weighting, and alternative sampling strategies.
- Adjust the classification threshold according to the bank's recall and precision priorities.
- Evaluate the financial costs of false positives and false negatives.
- Improve model interpretability by identifying the variables that contribute most to default predictions.
- Test the methodology on newer data or another credit-risk dataset.
- Create an application that allows the model to be tested in a controlled real-world workflow.

References

Ly, T., Schuller, R., Simanic, K., and Singh, S. Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms. University of Windsor.

UCI Machine Learning Repository. Default of Credit Card Clients, dataset ID 350.
<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>